Solution Requirements

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Data Ingestion	System shall load rainfall/weather data from flood_dataset.xlsx for training	High
2	Model Training	System shall train Decision Tree, Random Forest, KNN and XGBoost and select the best by F1-score	High
3	Prediction API	System shall expose a POST /api/predict endpoint accepting 10 weather features and returning flood probability	High
4	Web Interface	System shall provide a responsive form to submit weather data and view results without page reload	High
5	Result Logging	System shall persist every prediction (weather data + result) to a database	Medium
6	History View	System shall display the most recent predictions in a table	Medium
7	Error Handling	System shall validate inputs and return clear error messages for missing/invalid fields	Medium
8	Model Insights	System shall display auto-generated charts (comparison, confusion matrix) on the home page	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric
1	Performance & Speed	Prediction API should respond quickly for a single input	Response < 2 seconds
2	Scalability	Backend should handle multiple concurrent form submissions	Handles typical classroom/demo load (10+ concurrent users)
3	Security & Data Privacy	Inputs are validated server-side; no secrets committed to source control	No hard-coded credentials in repo
4	Reliability & Availability	App should recover cleanly from a bad/missing model file	Returns HTTP 503 with a clear message if model isn't trained
5	Usability & Accessibility	Form and results should be usable on mobile and desktop	Responsive layout down to 360px width
6	Maintainability	Code should be modular (separate ML, DB, routes) for easy updates	Clear app/ package

			structure with single-responsibility files